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BITS ID: 2022AC05327

Section : AIML (Section-2)

ISM Assignment

A1.) Given Data Set : 4, 12, 8, 14, 11, 6, 7, 13, 13, 11, 11, 20, 5, 19, 10, 15, 24, 7, 29, 6

$\therefore$ Count = 20

* Mean = $\bar{x} = \frac{\sum x_i}{N} = \frac{\text{Sum of data set}}{20} = \frac{245}{20} = \frac{49}{4} = 12.25$

* Median = (for even count $\therefore$ Count = 20)

$$= \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ value} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ value}}{2}$$

Arranging data set from least to greatest,

4, 5, 6, 6, 7, 7, 8, 10, 11, 11, 11, 12, 13, 13, 14, 15, 19, 20, 24, 29

$\therefore$ Middle data set = 10th data observation

$\therefore$ Median = $\frac{10^{\text{th}} \text{ observation} + 11^{\text{th}} \text{ observation}}{2}$

$$= \frac{11 + 11}{2} = 11$$

* Mode = Most frequency obs. in data set = 11 (appears thrice)

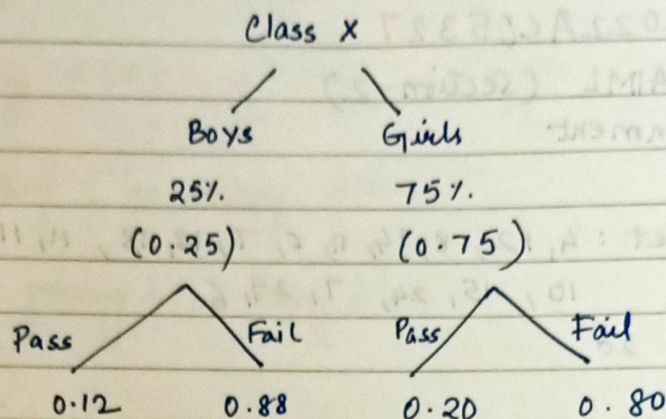
* Five-Point Summary :

- Min = 4
 - Max = 29
 - Median = 11
 - $IQ_1 = 7$
 - $IQ_3 = 14.5$
- $IQ_1 = \text{Median}(4, 5, 6, 6, 7, 7, 8, 10, 11, 11) = 7$
- $IQ_3 = \text{Median}(11, 12, 13, 13, 14, 15, 19, 20, 24, 29) = \frac{(14 + 15)}{2} = \frac{29}{2} = 14.5$

Box Plot :

Min	IQ_1	Median	IQ_3	Max
4	7	11	14.5	29

A.2>



$$\text{Now, } P(X) = P(B)P\left(\frac{X}{B}\right) + P(G)P\left(\frac{X}{G}\right)$$

$$\text{where } P(B) = 0.25 \quad P(G) = 0.75$$

$$P\left(\frac{X}{B}\right) = 0.12 \quad P\left(\frac{X}{G}\right) = 0.20$$

$P(X)$ = selected students passed in Maths

$$\begin{aligned} \Rightarrow P(X) &= (0.25)(0.12) + (0.75)(0.20) \\ &= 0.03 + 0.15 \\ &= 0.18 \end{aligned}$$

Hence, probability of students passed in Maths = 0.18

A.3>

Type I = 2 boxes

Type II = 3 boxes

$\therefore$ Total = 5 boxes

$$\Rightarrow P(\text{Type I box}) = \frac{2}{5}, \quad P(\text{Type II box}) = \frac{3}{5}$$

$$\begin{aligned} P(\text{White ball / Type I}) &= B_1 + B_2 \\ &= \frac{7}{20} + \frac{7}{20} \\ &= \frac{14}{20} = \frac{7}{10} \end{aligned}$$

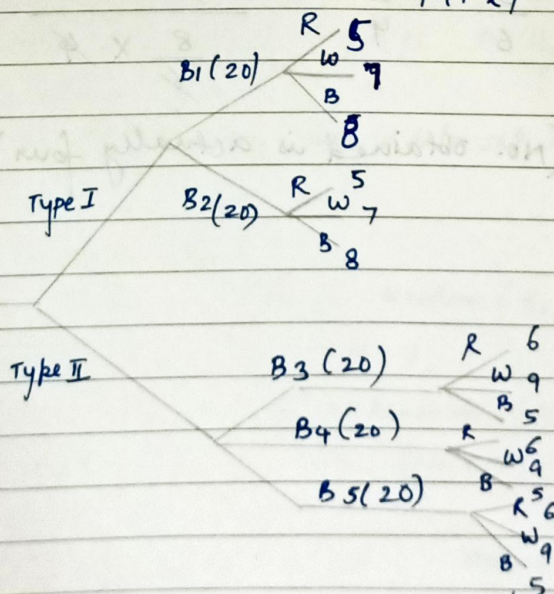
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$$\begin{aligned}
 P(\text{White ball / Type II}) &= B_1 + B_2 + B_3 \\
 &= \frac{9}{20} + \frac{9}{20} + \frac{9}{20} \\
 &= \frac{27}{60} = \frac{9}{20}
 \end{aligned}$$

According to Baye's Theorem,

$$P(\text{Type I / White Ball}) = \frac{P(\text{Type I}) \times P(\text{White Ball / Type I})}{[P(\text{Type I}) \times P(\text{White Ball / Type I}) + P(\text{Type II}) \times P(\text{White Ball / Type II})]}$$

$$\begin{aligned}
 \therefore P(\text{Type I / White Ball}) &= \frac{\frac{2}{5} \times \frac{7}{20}}{\frac{2}{5} \times \frac{7}{20} + \frac{3}{5} \times \frac{9}{20}} \\
 &= \frac{\frac{1}{5} \times \frac{1}{20} [2 \times 7]}{\frac{1}{5} \times \frac{1}{20} [(2 \times 7) + (3 \times 9)]} \\
 &= \frac{14}{14 + 27} = \frac{14}{41}
 \end{aligned}$$



$\therefore$ Probability of Type I ball such that it's white = $14/41$

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$$A4.) P(\text{Telling the truth}) = \frac{3}{4}$$

$$P(\text{Telling a lie}) = \frac{1}{4}$$

$$P(\text{Getting 4 on rolling dice}) = \frac{1}{6}$$

$$P(\text{Not getting 4 on rolling dice}) = \frac{5}{6}$$

$$\therefore P\left(\frac{\text{No. is 4 on rolling dice}}{\text{Telling truth}}\right)$$

$$= P(\text{Getting no. 4}) \cdot P\left(\frac{\text{Telling a truth}}{\text{Getting no. 4}}\right)$$

$$\left[P\left(\frac{\text{Getting no. 4}}{\text{Telling truth}}\right) \times P(\text{Telling truth}) \right] + \left[P(\text{Telling lie}) \times P\left(\frac{\text{Getting no. 4}}{\text{Telling lie}}\right) \right]$$

$$= \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{\frac{3}{8}}{\frac{3}{8} + \frac{5}{24}} = \frac{3}{8}$$

$$\therefore P(\text{No. obtained is actually four}) = \frac{3}{8}$$